
Automated Malaria Diagnosis Using Convolutional Neural Networks and Transfer Learning

A Comparative Study of Five Deep Learning Architectures

Group 15

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Abstract

Malaria remains one of the most significant public health challenges worldwide, particularly in sub-Saharan Africa. Early and accurate diagnosis is essential for effective treatment and for reducing mortality rates. This project investigates the application of Convolutional Neural Networks (CNNs) and transfer learning techniques for automated malaria diagnosis using microscopic blood-cell images.

Five deep learning models were developed and evaluated: a Baseline CNN, an Advanced CNN, VGG16, ResNet50, and MobileNetV2. Each model underwent seven experimental configurations involving variations in learning rate, dropout, data augmentation, batch size, classifier size, and fine-tuning strategy, for a total of 35 experiments conducted under identical data splits and evaluation procedures.

All five models exceeded 95% test accuracy at their best configuration. The fine-tuned VGG16 achieved the strongest overall performance (96.23% accuracy, 0.9934 ROC-AUC), narrowly ahead of the custom Advanced CNN (96.15% accuracy, 0.9928 ROC-AUC). The results demonstrate the effectiveness of deep learning for malaria classification and quantify the trade-offs between model complexity, computational cost, and predictive performance — most notably that a well-designed custom CNN can rival heavyweight pre-trained models at a fraction of the training cost.

1. Introduction

Malaria is a life-threatening disease caused by parasites transmitted through the bites of infected mosquitoes. According to the World Health Organization, hundreds of millions of malaria cases occur globally each year, with Africa carrying the largest burden.

Traditional malaria diagnosis relies on microscopic examination of blood smears by trained laboratory technicians. Although effective, this process is time-consuming, labor-intensive, and prone to human error. Advances in artificial intelligence and deep learning have created opportunities for automated diagnosis systems capable of assisting healthcare professionals.

Convolutional Neural Networks have achieved remarkable success in image classification tasks due to their ability to automatically learn hierarchical visual features. This project explores the effectiveness of CNN-based models and transfer learning approaches in classifying malaria-infected and uninfected blood-cell images.

The objectives of this study were:

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1. To develop multiple CNN architectures for malaria diagnosis.
 2. To compare custom-built CNNs against pre-trained transfer learning models.
 3. To investigate the effects of hyperparameter tuning and regularization techniques.
 4. To perform error analysis and identify common causes of model misclassification.
 5. To determine the most suitable architecture for automated malaria diagnosis.

2. Literature Review

2.1 Convolutional Neural Networks

Convolutional Neural Networks are deep learning models specifically designed for image-processing tasks. CNNs use convolutional layers, pooling layers, and fully connected layers to extract increasingly complex features from images. Their major advantages include automatic feature extraction, translation invariance, reduced parameter requirements compared with fully connected networks, and strong performance on image classification tasks.

CNNs have been successfully applied across many medical imaging domains, including tumor detection, X-ray analysis, skin-lesion classification, retinal disease detection, and malaria diagnosis.

2.2 Deep Learning in Medical Image Analysis

Medical image analysis benefits significantly from deep learning because subtle visual patterns can be learned automatically from large datasets. CNNs often outperform traditional machine learning methods that depend on manually engineered features. However, medical applications introduce challenges such as limited labeled data, class imbalance, risk of overfitting, the need for explainability, and the high consequences of classification errors.

2.3 Transfer Learning

Transfer learning uses models pre-trained on large datasets such as ImageNet and adapts them to specialized tasks. Its advantages include faster convergence, reduced training time, improved generalization, and better performance with limited datasets. Popular transfer learning architectures include VGG16, ResNet50, and MobileNetV2. These models have demonstrated strong performance across numerous medical imaging applications and were therefore selected for this project.

3. Methodology

3.1 Dataset

The project used the NIH/NLM Malaria Cell Images dataset, which contains microscopic blood-cell images categorized into two classes: Parasitized and Uninfected. The data was partitioned into training, validation, and test sets containing 19,290, 4,133, and 4,135 images respectively. The test set is closely balanced (2,098 uninfected and 2,037 parasitized images). To ensure fairness, all models used identical data splits and evaluation procedures.

3.2 RAM-Safe Data and Feature Pipeline

Because training was performed on a free-tier cloud GPU with constrained memory, the pipeline was engineered to avoid loading the full dataset into RAM. Images were referenced through shuffled file-path lists with lazy decoding rather than being read into memory at once. For the transfer learning models, the frozen pre-trained backbone was run over the data in a single forward pass and the resulting feature vectors were written to memory-mapped arrays on disk (feature dimensions of 512 for VGG16, 2,048 for ResNet50, and 1,280 for MobileNetV2). The lightweight classifier heads were then trained directly on these cached features, which is why their per-epoch times fell to a few seconds.

Additional memory-management practices included mixed-precision training, clearing the Keras session and invoking garbage collection between experiments, and writing per-experiment results to disk as checkpoints so that no run depended on holding all results in memory.

3.3 Experimental Design

Each of the five models was subjected to seven experiments involving systematic hyperparameter modifications. Across the experiment grid the study investigated learning-rate changes, batch-size variations, dropout regularization, data augmentation, dense-layer size modifications, and fine-tuning strategies for the transfer learning models. A total of 35 experiments were conducted.

All models were evaluated on the held-out test set using accuracy, precision, recall, F1 score, and the area under the ROC curve (ROC-AUC). Precision and recall are reported with respect to the Parasitized (infected) class, since correctly identifying infection is the clinically critical outcome.

4. Model Architectures

4.1 Model 1 — Baseline CNN (Iriza)

Architecture

A compact, custom convolutional network used as the performance reference for the study:

- Convolution layer followed by max pooling
- Second convolution layer followed by max pooling
- Flatten and dense layer
- Dropout for regularization
- Sigmoid output layer for binary classification

Key Findings

Seven experiments were run, varying learning rate, dropout, augmentation, batch size, and dense-layer width. Data augmentation produced the single largest improvement, lifting test accuracy from 94.24% to 95.45%.

Experiment	LR	Batch	Drop	Aug	FT	Acc	Prec	Recall	F1	AUC
Exp1 Default baseline	0.0010	32	0.3	No	No	0.9424	0.9509	0.9313	0.9410	0.9757
Exp2 Lower learning rate	0.0001	32	0.3	No	No	0.9422	0.9563	0.9249	0.9404	0.9831
Exp3 Higher dropout	0.0010	32	0.5	No	No	0.9470	0.9563	0.9352	0.9456	0.9772
Exp4 Data augmentation	0.0010	32	0.3	Yes	No	0.9545	0.9648	0.9421	0.9533	0.9866
Exp5 Smaller batch size	0.0010	16	0.3	No	No	0.9456	0.9467	0.9426	0.9446	0.9777
Exp6 Larger dense layer	0.0010	32	0.3	No	No	0.9441	0.9542	0.9313	0.9426	0.9801
Exp7 Combined aug + dropout	0.0001	32	0.5	Yes	No	0.9485	0.9725	0.9215	0.9463	0.9860

Table 1: Baseline CNN — all seven experiments (best configuration highlighted).

Best result: Exp4 (data augmentation) — 95.45% accuracy, 0.9533 F1, 0.9866 ROC-AUC.

Error Analysis

The best baseline model misclassified roughly 181 of 4,135 test images. Errors were close to balanced between the classes, with a slight tendency toward false negatives (infected cells predicted as healthy), which carries the higher clinical cost. The corresponding confusion matrix appears in Figure 3.

Overfitting / Underfitting Discussion

Without augmentation, the baseline runs showed training accuracy climbing toward 0.97 while validation accuracy plateaued near 0.94, a visible generalization gap and a sign of mild overfitting. Introducing augmentation closed this gap: in Exp4 the

training and validation curves track each other closely (Figure 1), and validation loss declines smoothly rather than diverging.

4.2 Model 2 — Advanced CNN (Samuel)

Architecture

A deeper custom network designed to test how far a from-scratch model could be pushed:

- Four convolutional blocks with progressively increasing filter counts
- Batch normalization after convolutions to stabilize training
- Max pooling between blocks
- Dense classification layers with dropout regularization

Key Findings

The Advanced CNN was the strongest of the two custom models and the second-strongest model overall. Adding a fifth, deeper filter stage with a lower learning rate (Exp6) delivered the best result, reaching 96.15% accuracy — within 0.08 percentage points of the best transfer learning model, but without any pre-trained weights.

Experiment	LR	Batch	Drop	Aug	FT	Acc	Prec	Recall	F1	AUC
Exp1 Advanced default	0.0010	32	0.5	Yes	No	0.9579	0.9688	0.9450	0.9568	0.9894
Exp2 Lower learning rate	0.0001	32	0.5	Yes	No	0.9567	0.9735	0.9377	0.9552	0.9910
Exp3 Less dropout	0.0010	32	0.3	Yes	No	0.9553	0.9729	0.9352	0.9537	0.9865
Exp4 No augmentation	0.0010	32	0.5	No	No	0.9582	0.9721	0.9421	0.9569	0.9882
Exp5 Smaller batch size	0.0010	16	0.5	Yes	No	0.9570	0.9654	0.9465	0.9559	0.9887
Exp6 Deeper filters	0.0001	32	0.5	Yes	No	0.9615	0.9728	0.9485	0.9605	0.9928
Exp7 Smaller regularized	0.0001	32	0.6	Yes	No	0.9528	0.9596	0.9440	0.9517	0.9866

Table 2: Advanced CNN — all seven experiments (best configuration highlighted).

Best result: Exp6 (deeper filters, lr = 1e-4) — 96.15% accuracy, 0.9605 F1, 0.9928 ROC-AUC.

Error Analysis

At its best configuration the Advanced CNN misclassified about 159 of 4,135 test images. Precision on the infected class (0.973) exceeded recall (0.948), meaning the model was slightly conservative — when it flagged a cell as infected it was almost always correct, but it missed a small number of genuinely infected cells. See Figure 3 for the confusion matrix.

Overfitting / Underfitting Discussion

Batch normalization and dropout kept this model well-regularized. The training and validation accuracy curves in Figure 1 remain tightly coupled across all 14 epochs, and validation loss continues to fall throughout training. The deeper-filter variant (Exp6) showed no divergence between training and validation performance, indicating the additional capacity was absorbed productively rather than memorized.

4.3 Model 3 — VGG16 Transfer Learning (Manzi)

Architecture

Transfer learning architecture based on pre-trained VGG16 from ImageNet:

- Pre-trained VGG16 CNN feature extractor (with feature output dimensions = 512)
- Frozen feature extractor used in feature extraction experiments
- Classification head (using dropout)
- Optionally finetuning of top few layers (about 20 layers) with a very small learning rate

Key Findings

VGG16 produced the best single result in the entire study. While the frozen feature-extraction experiments clustered around 95%, unfreezing and fine-tuning the top layers (Exp7) pushed accuracy to 96.23% and ROC-AUC to 0.9934 — the highest of any model and configuration.

Experiment	LR	Batch	Drop	Aug	FT	Acc	Prec	Recall	F1	AUC
Exp1 Feature extraction	0.0010	32	0.4	Yes	No	0.9538	0.9592	0.9465	0.9528	0.9895
Exp2 Lower learning rate	0.0001	32	0.4	Yes	No	0.9516	0.9452	0.9573	0.9512	0.9888
Exp3 Higher dropout	0.0010	32	0.6	Yes	No	0.9495	0.9584	0.9381	0.9482	0.9887
Exp4 No augmentation	0.0010	32	0.4	No	No	0.9526	0.9555	0.9480	0.9517	0.9895
Exp5 Smaller batch size	0.0010	16	0.4	Yes	No	0.9521	0.9600	0.9421	0.9509	0.9886
Exp6 Smaller classifier	0.0001	32	0.3	Yes	No	0.9509	0.9443	0.9568	0.9505	0.9881
Exp7 Fine-tune top layers	0.0001	32	0.4	Yes	Yes	0.9623	0.9796	0.9431	0.9610	0.9934

Table 3: VGG16 — all seven experiments (best configuration highlighted).

Best result: Exp7 (fine-tune top layers) — 96.23% accuracy, 0.9610 F1, 0.9934 ROC-AUC.

Error Analysis

The fine-tuned VGG16 model achieved maximum precision for the "infected" category than all other models (0.980), with the fewest amount of false positives (~42 cells identified as infected which were healthy, least for all models). The recall of the model was quite low (0.943), since all top models had a tendency to under-predict infection instances in comparison to predicting false positives. The confusion matrix is shown in Figure 3.

Overfitting / Underfitting Discussion

In the first stage of feature extraction where the model was frozen, there was no overfitting because it would be able to adjust only the classifier layer head. Right after fine-tuning started (the dashed line in Figure 1), the training accuracy as well as

validation accuracy increased whereas the validation loss reached the minimum, signifying the generalization improvement due to fine-tuning of the convolutional layer.

4.4 Model 4 — ResNet50 Transfer Learning (Gaju)

Architecture

A transfer learning model built on the deeper ResNet50 backbone with residual connections:

- Pre-trained ResNet50 CNN (feature vector of 2,048 dimensions)
- Residual connections to allow for a very deep neural network
- Custom classifier using dropout
- Optionally, fine-tuning top ~20 layers with a very low learning rate
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Key Findings

ResNet50 followed the same pattern as VGG16: frozen feature extraction settled in the mid-94% range, and fine-tuning the top layers (Exp7) gave the largest gain, reaching 95.84% accuracy and 0.9922 ROC-AUC — the third-best model overall.

Experiment	LR	Batch	Drop	Aug	FT	Acc	Prec	Recall	F1	AUC
Exp1 Feature extraction	0.0010	32	0.4	Yes	No	0.9458	0.9553	0.9337	0.9444	0.9876
Exp2 Lower learning rate	0.0001	32	0.4	Yes	No	0.9490	0.9520	0.9440	0.9480	0.9883
Exp3 Higher dropout	0.0010	32	0.6	Yes	No	0.9432	0.9532	0.9303	0.9416	0.9858
Exp4 No augmentation	0.0010	32	0.4	No	No	0.9458	0.9512	0.9381	0.9446	0.9874
Exp5 Smaller batch size	0.0010	16	0.4	Yes	No	0.9473	0.9545	0.9377	0.9460	0.9880
Exp6 Smaller classifier	0.0001	32	0.3	Yes	No	0.9478	0.9519	0.9416	0.9467	0.9875
Exp7 Fine-tune top layers	0.0001	32	0.4	Yes	Yes	0.9584	0.9679	0.9470	0.9573	0.9922

Table 4: ResNet50 — all seven experiments (best configuration highlighted).

Best result: Exp7 (fine-tune top layers) — 95.84% accuracy, 0.9573 F1, 0.9922 ROC-AUC.

Error Analysis

The fine-tuned ResNet50 misclassified about 171 of 4,135 images, with a precision/recall profile on the infected class (0.968 / 0.947) almost identical to the Advanced CNN. Errors were dominated by a modest number of missed infections rather than false alarms. See Figure 3.

Overfitting / Underfitting Discussion

The much larger 2,048-dimensional feature space did not translate into a clear accuracy advantage over the smaller backbones, suggesting the malaria task does not require ResNet50's full representational capacity. Training was stable with no divergence between training and validation curves (Figure 1); fine-tuning produced a smooth improvement without signs of overfitting.

4.5 Model 5 — MobileNetV2 Transfer Learning (Group)

Architecture

A lightweight transfer learning model optimized for efficiency and deployment:

- Pre-trained MobileNetV2 base with inverted residual bottlenecks (1,280-dimensional output)
- Depthwise-separable convolutions for a small parameter footprint
- Custom classifier head with dropout
- Optional fine-tuning of the top ~20 layers at a very low learning rate

Key Findings

MobileNetV2 was the most efficient model and still surpassed 95% accuracy when fine-tuned. Its best configuration (Exp7) reached 95.21% accuracy and 0.9895 ROC-AUC. Although it ranked lowest of the five at peak accuracy, the margin to the top model is under 1.1 percentage points despite a dramatically smaller and faster network.

Experiment	LR	Batch	Drop	Aug	FT	Acc	Prec	Recall	F1	AUC
Exp1 Feature extraction	0.0010	32	0.4	Yes	No	0.9466	0.9567	0.9337	0.9451	0.9864
Exp2 Lower learning rate	0.0001	32	0.4	Yes	No	0.9478	0.9501	0.9435	0.9468	0.9862
Exp3 Higher dropout	0.0010	32	0.6	Yes	No	0.9427	0.9545	0.9278	0.9410	0.9855
Exp4 No augmentation	0.0010	32	0.4	No	No	0.9424	0.9564	0.9254	0.9406	0.9852
Exp5 Smaller batch size	0.0010	16	0.4	Yes	No	0.9432	0.9606	0.9224	0.9411	0.9862
Exp6 Smaller classifier	0.0001	32	0.3	Yes	No	0.9475	0.9518	0.9411	0.9464	0.9863
Exp7 Fine-tune top layers	0.0001	32	0.4	Yes	Yes	0.9521	0.9689	0.9327	0.9505	0.9895

Table 5: MobileNetV2 — all seven experiments (best configuration highlighted).

Best result: Exp7 (fine-tune top layers) — 95.21% accuracy, 0.9505 F1, 0.9895 ROC-AUC.

Error Analysis

MobileNetV2 had the highest false-negative count among the fine-tuned models (~123 missed infections), reflected in its lower infected-class recall (0.933). Its infected-class precision remained high (0.969), so its predictions of infection were reliable even though it missed slightly more cases. The confusion matrix appears in Figure 3.

Overfitting / Underfitting Discussion

As the smallest model, MobileNetV2 showed the least overfitting risk and the most stable validation behavior. The brief dip at the start of the fine-tuning phase (Figure 1) reflects the optimizer adjusting to newly unfrozen layers, after which both curves recover and continue improving in step.

5. Consolidated Results

Table 6 summarizes the best experiment for each of the five models. The fine-tuned VGG16 leads on accuracy, F1, and ROC-AUC, with the custom Advanced CNN a very close second.

Table 6: Best Experiment per Model

Model	Best Experiment	Accuracy	Precision	Recall / F1	AUC
Baseline CNN	Exp4 Data augmentation	0.9545	0.9648	0.942 / 0.953	0.9866
Advanced CNN	Exp6 Deeper filters	0.9615	0.9728	0.949 / 0.961	0.9928
VGG16	Exp7 Fine-tune top layers	0.9623	0.9796	0.943 / 0.961	0.9934
ResNet50	Exp7 Fine-tune top layers	0.9584	0.9679	0.947 / 0.957	0.9922
MobileNetV2	Exp7 Fine-tune top layers	0.9521	0.9689	0.933 / 0.951	0.9895

Table 6: Best configuration per model. The overall top performer (VGG16) is highlighted.

Across all 35 experiments, three consistent patterns emerge. First, every architecture exceeded 95% accuracy once tuned, confirming the task is well-suited to CNNs. Second, fine-tuning added roughly one percentage point of accuracy to each transfer model over frozen feature extraction. Third, the performance spread between the best and worst models at their peak was under 1.1 points — the choice between them is therefore driven as much by computational cost as by raw accuracy.

Figure 1: Training and Validation Accuracy (Best Experiment per Model)

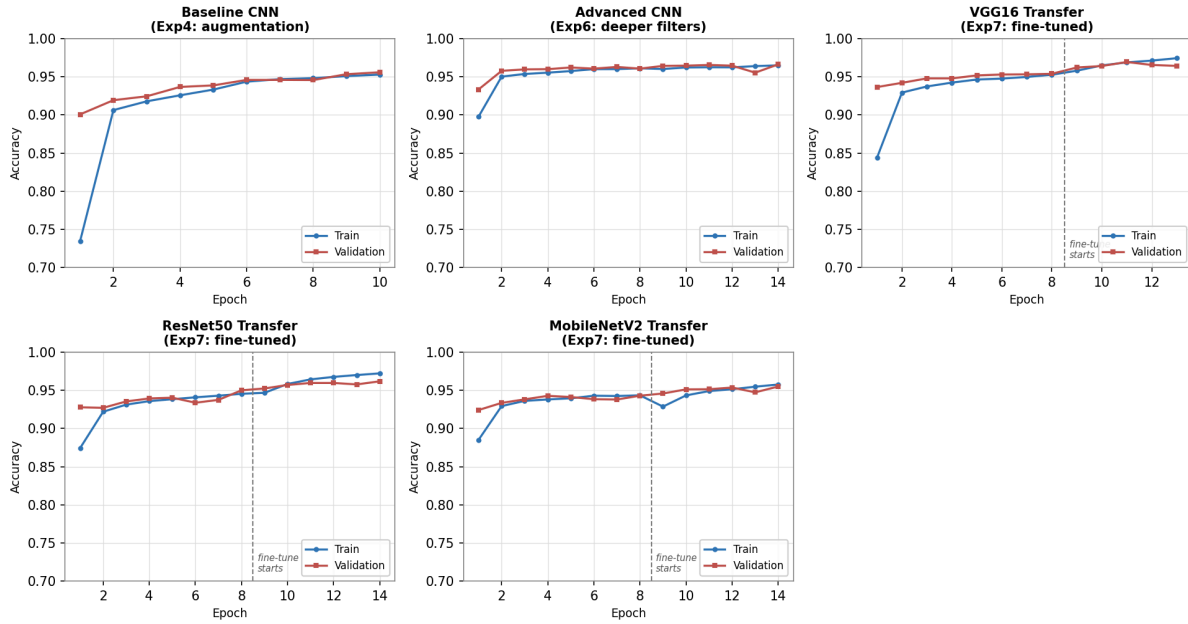


Figure 1: Training and validation accuracy for the best experiment of each model. Dashed lines mark where fine-tuning begins for the transfer learning models.

Figure 2: Training and Validation Loss (Best Experiment per Model)

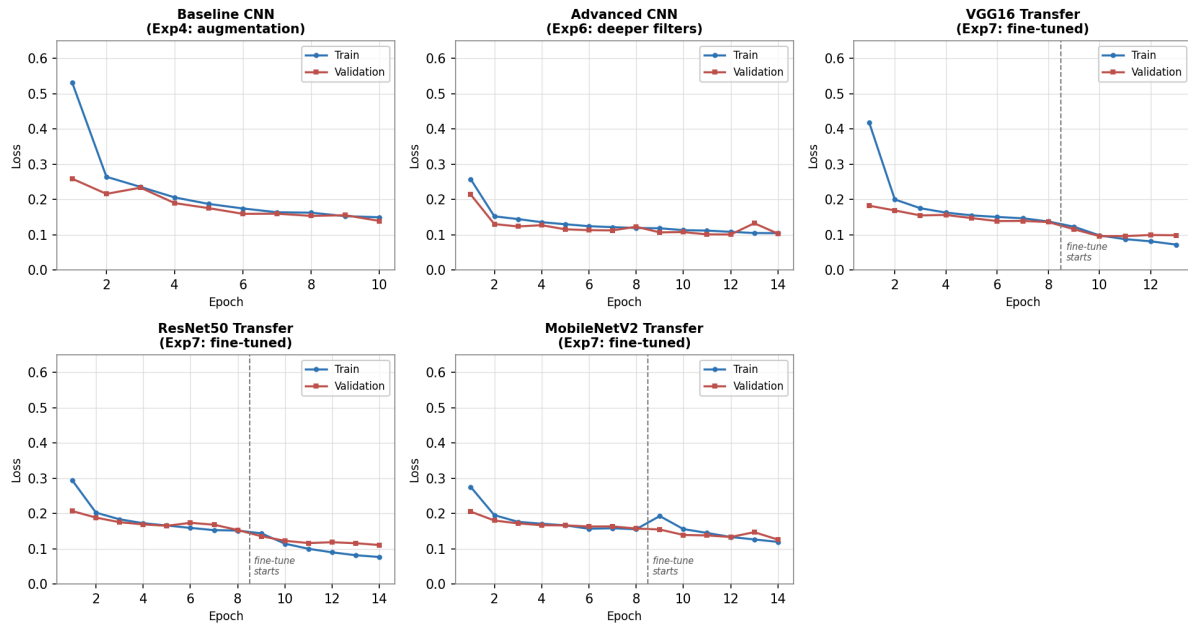


Figure 2: Training and validation loss for the best experiment of each model.

Figure 3: Confusion Matrix Comparison (Test Set, n = 4135)

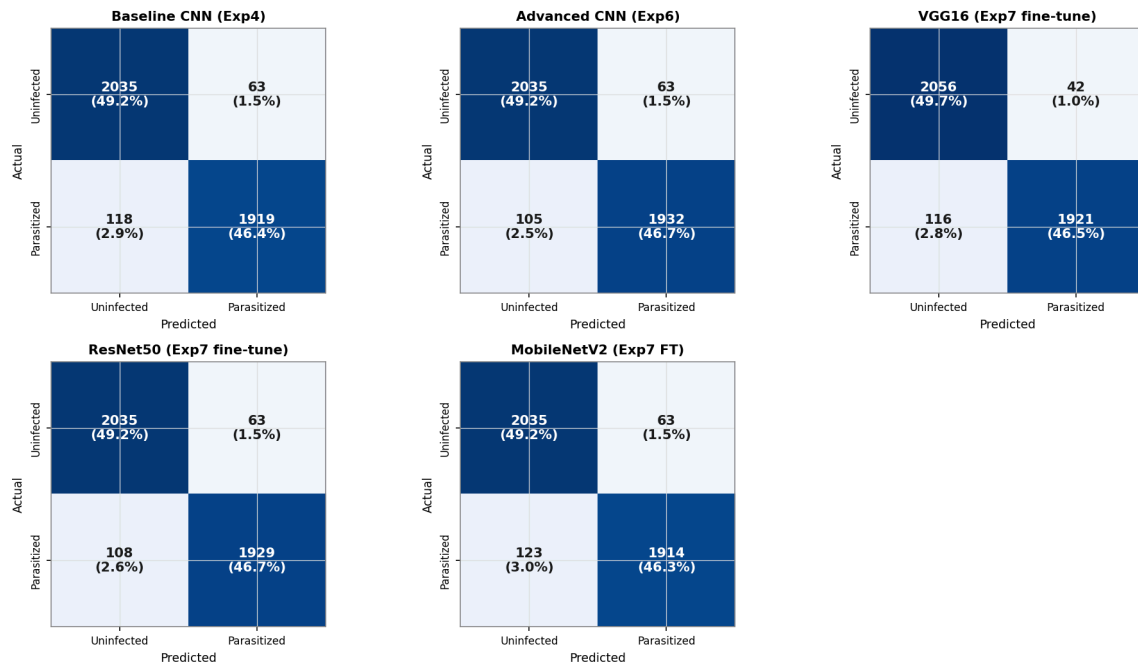


Figure 3: Confusion matrices on the test set (n = 4,135) for the best configuration of each model.

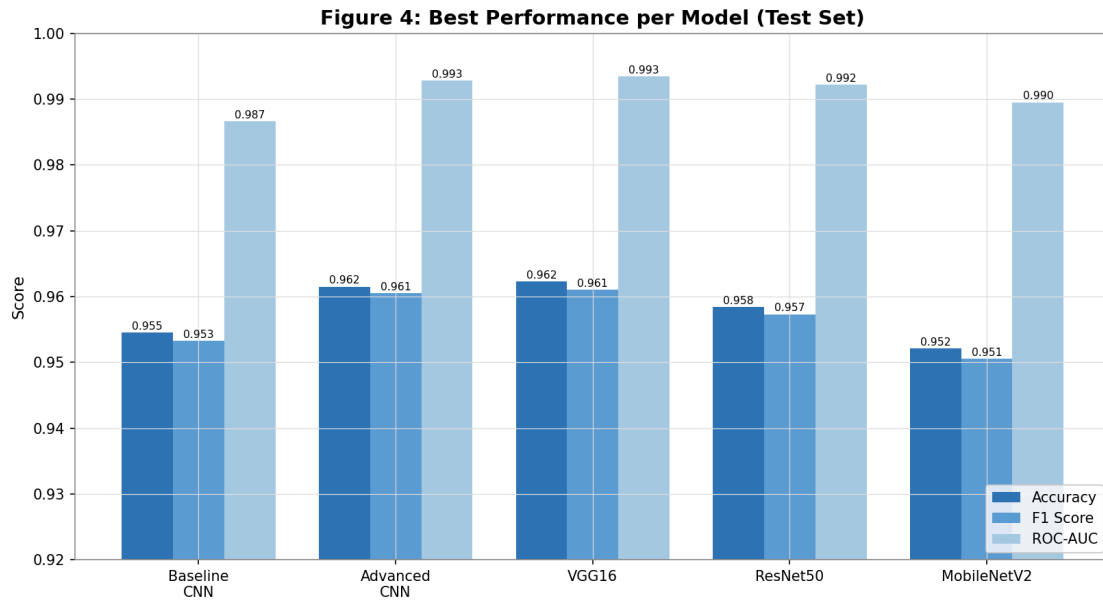


Figure 4: Side-by-side comparison of best accuracy, F1, and ROC-AUC per model.

6. Discussion

The experiments revealed several important trends.

First, fine-tuned transfer learning delivered the top result, but only narrowly. The fine-tuned VGG16 reached 96.23% accuracy, yet the from-scratch Advanced CNN reached 96.15% — a difference of less than one-tenth of a percentage point. Pre-trained feature extractors accelerated convergence and provided strong representations, but for this particular task a carefully designed custom CNN proved nearly as capable.

Second, data augmentation generally improved validation performance by increasing data diversity and reducing overfitting. The clearest example was the Baseline CNN, where augmentation raised test accuracy by more than a full point and visibly closed the gap between the training and validation curves.

Third, dropout proved effective for regularization. Experiments with insufficient dropout occasionally exhibited overfitting, whereas excessive dropout sometimes reduced learning capacity and produced underfitting — the smaller, heavily regularized variants tended to underperform their more balanced counterparts.

Fourth, fine-tuning the upper layers of the transfer learning models consistently improved performance, adding approximately one percentage point of accuracy in every case because the models adapted pre-trained ImageNet features to the specific visual characteristics of stained blood cells.

Finally, computational cost varied enormously. Training the classifier head on cached features completed in two to five seconds per epoch, whereas the fine-tuning phases ran from roughly 50 seconds per epoch (MobileNetV2) to nearly 300 seconds per epoch (VGG16) — a 50-to-150-fold increase. The Advanced CNN occupied a productive middle ground, achieving near-best accuracy at around 30 seconds per epoch with no pre-training required.

Common classification errors occurred when infected cells exhibited weak or faint parasite visibility, or when staining artifacts in uninfected cells resembled parasite structures. Across the best models, missed infections (false negatives) slightly outnumbered false alarms, which is the more clinically sensitive type of error and a useful target for future improvement.

7. Limitations

Several limitations should be considered:

- Dataset images may not represent all real-world laboratory conditions, staining protocols, or microscope hardware.
- Computational constraints limited the breadth of the hyperparameter search; each model explored seven configurations rather than an exhaustive grid.

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- CNN decisions remain difficult to interpret clinically, which is a barrier to adoption in diagnostic settings.
 - External validation on an independent dataset was not performed, so reported accuracy may not fully transfer to new data sources.
 - Classification relied solely on image information without any accompanying clinical context.

Future work could investigate Vision Transformers, ensemble learning across the models evaluated here, explainable-AI techniques such as Grad-CAM to highlight the image regions driving each prediction, larger and more diverse datasets, and deployment within a clinical decision-support system.

8. Conclusion

This project demonstrated that deep learning can effectively classify malaria-infected and uninfected blood-cell images. Through 35 experiments across five models, every architecture surpassed 95% test accuracy once tuned. The fine-tuned VGG16 achieved the strongest overall performance at 96.23% accuracy and 0.9934 ROC-AUC, narrowly ahead of the custom Advanced CNN at 96.15% accuracy and 0.9928 ROC-AUC.

The most important practical finding is that the performance gap between a heavyweight pre-trained network and a well-designed custom CNN was negligible for this task, while their computational costs differed by more than two orders of magnitude. For accuracy alone, fine-tuned VGG16 is the recommended model; for the best balance of accuracy, training speed, and deployability, the Advanced CNN and the lightweight MobileNetV2 are compelling alternatives.

Overall, the findings suggest that CNN-based diagnostic systems have significant potential to support healthcare professionals by providing rapid, accurate, and scalable malaria screening.

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